HOW CAN DUNDEE'S GREENSPACES CONTRIBUTE TO WELLBEING?

EXPLORING HOW DUNDEE'S GREENSPACES IMPACT RESIDENT WELLBEING AND HOW BIODIVERSITY LOSS AFFECTS GREENSPACE HEALTH.

Sustainable Development Goals

SDG 3: Ensure healthy lives and promote well-being for all at all ages. Connect with nature, enjoy your local greenspace. Walk! It's good for the mind, body and planet. Take regular exercise, find a sport or activity you enjoy.

SDG 11: Make cities and human settlements inclusive, safe, resilient and sustainable. Take care of public spaces. Organise clean ups, plant trees, renovate.

SDG 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss. Clean your local parks and forests. Protect biodiversity and natural habitats.

Earth Charter

Respect and Care for the Community of Life: Care for the community of life with understanding, compassion, and love. Accept that with the right to own, manage, and use natural resources comes the duty to prevent environmental harm and to protect the rights of people.

Ecological Integrity: Protect and restore the integrity of Earth's ecological systems, with special concern for biological diversity and the natural processes that sustain life. Adopt at all levels sustainable development plans and regulations that make environmental conservation and rehabilitation integral to all development initiatives. Promote the recovery of endangered species and ecosystems.

Inner Development Goals

Being: Connection to nature to encourage presence, mindfulness, and collectedness. Openness & Learning Mindset. Self-Awareness. Presence.

Relating: Caring for others and the environment we live in. Appreciation. Connectedness. Empathy & Compassion.

Collaborating: Greenspaces increase social interaction and collaboration. Co-creation Skills. Communication Skills. Mobilisation Skills.

AIMS AND OBJECTIVES

Explore the relationship between greenspaces and wellbeing metrics.

Assess current biodiversity status and highlight restoration plans for natural flora and fauna.

Understand possible solutions to Dundee's socioeconomic challenges.

Intended audience:

- Dundee residents interested in wellbeing factors
- Dundee City Council planning, environmental, and building departments for development considerations

WHAT CAN GREENSPACES DO FOR DUNDEE?

92% OF RESPONDENTS TO THE GREENSPACE USE AND ATTITUDES SURVEY AGREE THAT IT IS IMPORTANT TO HAVE GREENSPACE IN THEIR LOCAL AREA.

According to the Third State of Scotland's Greenspace report (2018), greenspaces have shown to contribute to the quality of life in various ways:

Healthier communities: Mental and physical wellbeing, sport and activities, and healthy eating.

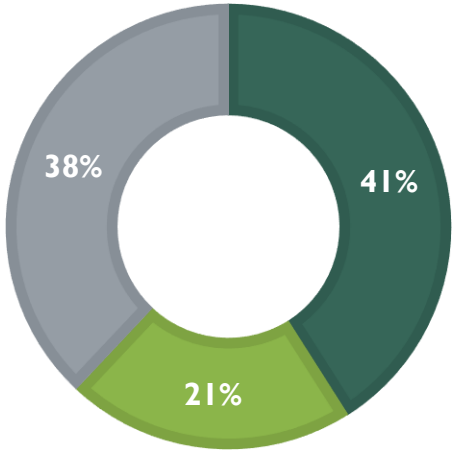
Greener communities: Growing local food, green networks and infrastructure, and supporting biodiversity.

Smarter communities: Environmental education and volunteering, skills and confidence.

Safer, stronger, communities: Reduced crime, and places to meet your community.

Wealthier and fairer communities: Attracts inward investment, green workplaces have healthier employees, places people want to live and work.

PERCENTAGE OF GREENSPACE AREA IN DUNDEE



FACT: **41%** of Dundee is publicly accessible greenspace – **62%** including private gardens, which is the highest percentage of publicly accessible greenspace in Scotland!

HOW CAN WE KEEP OUR GREENSPACES HEALTHY?

FROM THE AIR WE BREATHE AND THE WATER WE DRINK, TO THE PLACES WE GO FOR RECREATION AND OUR MENTAL WELLBEING; BIODIVERSITY AFFECTS US ALL

Greater levels of biodiversity helps to ensure the natural sustainability of all life, which is especially important in a time of global climate change. The State of Nature Report 2019 revealed that **41%** of UK species studied have shown strong or moderate declines since 1970, while **15%** of species assessed in the UK are threatened with extinction, and **2%** are already extinct.

Biodiversity Action Plan Objectives

- Promote sympathetic & sustainable management for ecosystem health.
- Ensure no net loss of habitat and to increase the distribution and connectivity of habitats.
- Find a balance between providing for biodiversity and providing public amenity.

QUALITY MATTERS: 40% of respondents to the Greenspace Use and Attitudes Survey (2017) agreed or strongly agreed that 'the quality of my local greenspace has reduced in the past 5 years' (up from 33% in 2011). This figure rises to 50% for respondents from the 15% most deprived areas (up from 35% in 2011).

Ecosystem Threats

- Human activity and management
- Development and fragmentation
- Climate change
- Invasive Non-native Species (INNS)
- Pollution
- Nutrient enrichment
- Over exploitation of resources

Ecosystem Benefits

Provisioning Services: Opportunity for foraging.

Regulating Services: Carbon sequestration and storage in soils, pollination, and flood management control.

Cultural Services: Recreation/tourism, spiritual/religious, aesthetic/inspiration.

Supporting Services: Soil formation, photosynthesis, and water recycling.

COMMUNITY INVOLVEMENT

50% OF RESPONDENTS TO THE GREENSPACE USE AND ATTITUDES SURVEY AGREED THEY WOULD LIKE MORE SAY IN HOW THEIR LOCAL GREENSPACE IS MANAGED.

Urban ReLeaf project

Aims to harness the collective input of Dundee's citizens to enhance the benefits derived from greenspaces and ensure they cater to the diverse needs of the community

Wee Forests project

Densley planted and fast growing, multi-layered native species rich woodland in urban areas, which provides benefits for long-term learning and play, biodiversity and local ecosystems services, and children's well-being.

Eden project

£130 million regeneration and green tourism initiative transforming Dundee's former gasworks site into a sustainability hub providing immersive biodiversity learning, world class horticulture, and cultural events.

Achievements from Biodiversity Duty Report (2021-2023)

Grassland managed for biodiversity

49 ha naturalised grassland

10 ha biodiversity grassland with **2 ha** improved with additional seeding

50,000 native bulbs planted

10,000 allium bulbs planted

3.4 ha existing native meadows

1 ha annual flower areas

3.3 ha native meadows in partnership with Eden

32,000 native trees planted

Trees and Woodland

32,000 native trees planted

3,200 trees surveyed for disease

Invasive Non-Native Species

40 ha of DCC owned areas with giant hogweed mapped with ongoing management

Community Engagement

4,150 hours of volunteering

45 public events

1800 children engaged

50 hours of STEM teacher training

10 Friends of Groups

169 Litter picks across the city by volunteers and Community groups

12 Its Your Neighbourhood groups

5 Community Anchor groups in Dundee for Upstream Battle on the Tay

Urban & Gardening

26 Community Gardens

550 Allotment Plots

45% reduction in herbicide usage by DCC since 2021

2 new pocket parks

100 amphibian ladders installed in roadside drains

DUNDEE'S SOCIOECONOMIC ISSUES

ECONOMIC DEPRIVATION INDICATORS MOSTLY SUGGESTED THAT THE LEVEL OF DEPRIVATION IN DUNDEE CITY WAS HIGHER THAN THE LEVEL FOR SCOTLAND OVERALL.

Social and Health Challenges

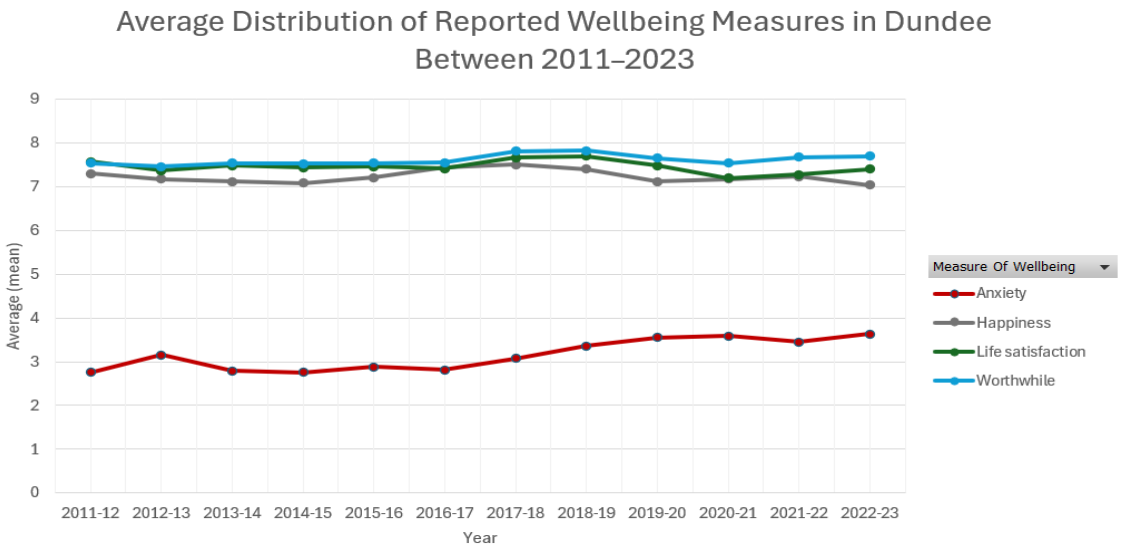
Millard et., al (2016) found that Dundee has higher significant social and health challenges:

- In 2012, the rate for prisoner population was 347/100,000, which was **103% higher** than the Scottish level of 171/100,000
- In 2012-2015 the rate for drug-related hospital stays was 221/100,000, which was **82% higher** than the Scottish level of 122/100,000.
- In 2014 the crude rate of crime rate was 55/1000, which was **35% higher** than the Scottish level of 40/1000.
- In 2014/15 the percentage of people prescribed medication for anxiety, depression or psychosis was **20% higher** than Scotland overall.

Research from Ogletree, et., al (2022) indicates that greenspaces can address some of these challenges in a beneficial way by increasing the amount of quality greenspace and improving accessibility for the entire community.

Wellbeing Reports

Data from the ONS (2023) shows that happiness, life satisfaction, and the feeling things done in life are worthwhile remain fairly stable between 2011-2023, however it also reveals a slight increasing trend in reported anxiety levels. Understanding contributing factors for this increase is important to ensure community wellbeing.



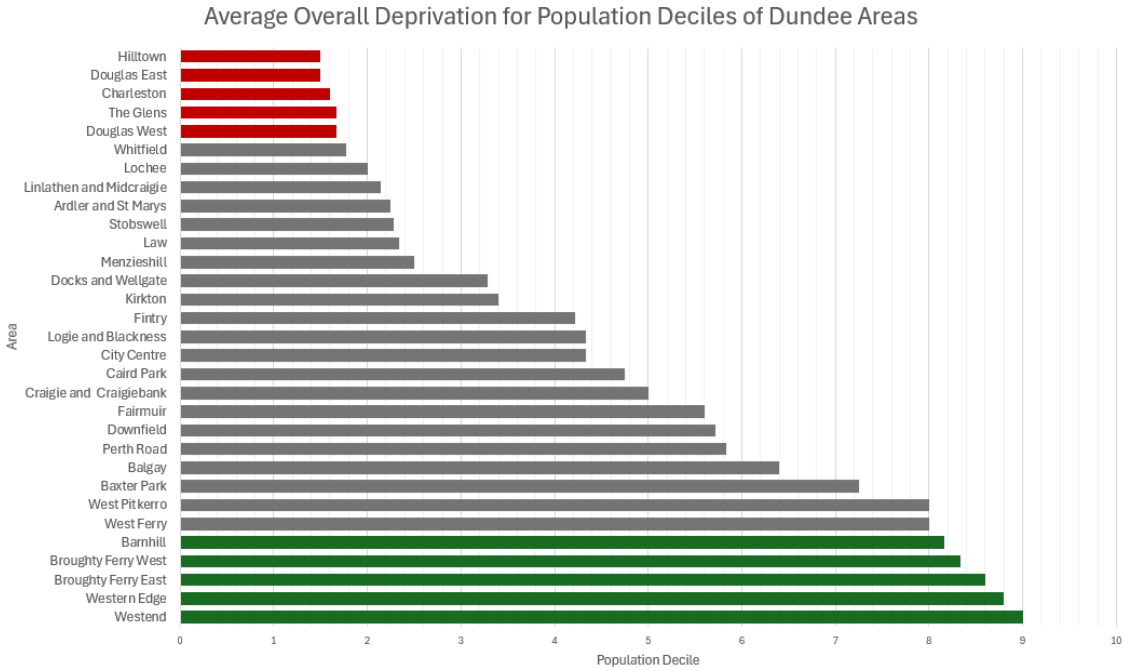
DOES PROXIMITY TO GREENSPACE CONTRIBUTE TO WELLBEING?

HALF (50%) OF RESPONDENTS IN THE 15% MOST DEPRIVED AREAS AGREED OR AGREED STRONGLY THAT 'THE QUALITY OF THEIR LOCAL GREENSPACE HAS BEEN REDUCED'. ?

Green Space Accessibility

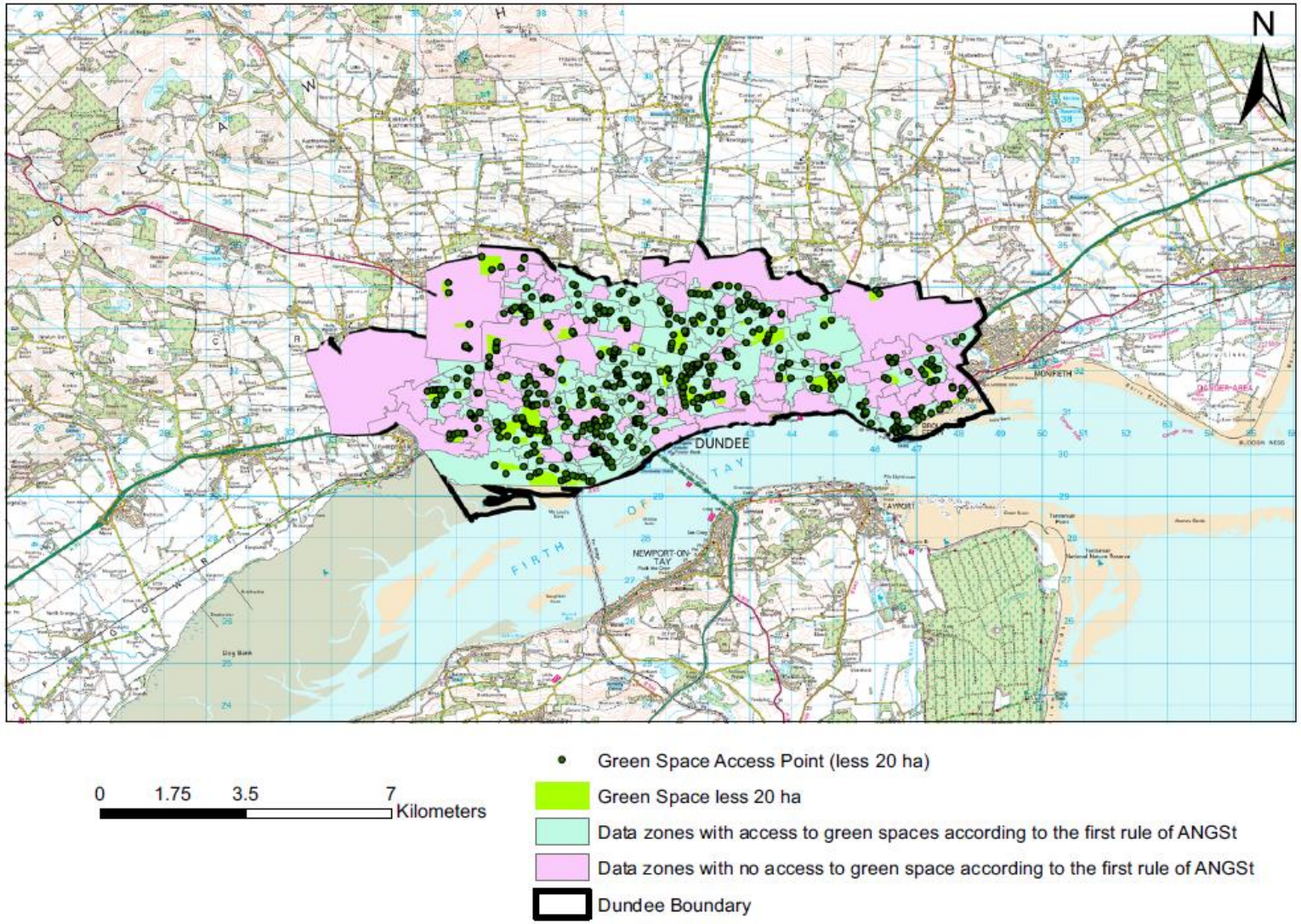
By Accessible Natural Greenspace Standards (ANGSt), Kmail and Onyango (2020) discovered that out of 188 data zones in Dundee:

- **57%** have access to greenspace according to the first requirement of ANGSt – at least one accessible 2 ha site, no more than 300 m (5 minutes walk) from home.
- **51.5%** have access to greenspace according to the second requirement of ANGSt – at least one accessible 20 ha site within 2 km from home.
- **46.80%** have access to greenspace according to the third rule of ANGSt – one accessible 100 ha site within 5 km of home.



Some of Dundee's most deprived zones have easily accessible greenspace, yet some of Dundee's least deprived zones do not.

Data zones that have access to green spaces according to the first requirement of ANGSt



Source: Kmail, A. B., and Onyango, V (2020) A GIS-based assessment of green space accessibility: case study of Dundee, Fig. 3.

SUMMARY

- There is research that indicates spending more time in nature positively influences wellbeing.
- There is no direct relationship or data that suggests more greenspaces alone will directly result in better wellbeing.
- Dundee has the highest percentage of publicly available greenspace in Scotland yet has substantial social and health challenges.
- People consistently report that the health and quality of their local greenspaces are declining.
- There are imminent threats to our ecosystem that benefits us all to address.

Conclusion

Dundee's priority should be to encourage people to engage with nature and to actively contribute to local community efforts which restore and enhance the health of our greenspaces. These efforts can not only strengthen local ecosystems but also improve overall wellbeing for everyone.

REFERENCES

Data

- LDP2019 Open Space contains GIS data for all public greenspace in Dundee.

  This data was obtained from Dundee Open Data portal. The source of this information is licenced under Crown copyright and database rights 2023 Ordnance Survey 100023371 and made available under the Open Government Licence 3.0.

- Personal well-being estimates by local authority contains estimates of life satisfaction across the UK [filtered to include only Dundee data].

  This data was obtained from the Office for National Statistics (ONS) under the Open Government Licence 3.0.

- Dundee City SIMD 2020 Social Deprivation data which uses 7 domains to measure deprivation over 188 zones in Dundee.

  This data was obtained from the Scottish Government and is publicly available.

- Play Area Sports and Recreation Playground contains data for recreational greenspace areas in Dundee.

  This data was obtained from Dundee Open Data portal and is licenced under Open Government Licence 3.0.

Reports and Other Resources

- Biodiversity Action Plan (2020)
- Biodiversity Duty Report (2021-2023)
- Eden Project
- Greenspace Use and Attitudes Survey (2017)
- Kmail, A. B., and Onyango, V (2020) A GIS-based assessment of green space accessibility: case study of Dundee.
- Millard A, McCartney G, MacKinnon A, Van Heelsum A, Gasiorowski A, Barkat S. (2016) *Dundee City Health and Wellbeing Profiles – key indicators and overview*
- Ogletree, S, Larson, L, Powell, RB, White, DL & Brownlee, MTJ (2022) *Urban greenspace linked to lower crime risk across 301 major U.S. cities*, Cities, vol. 131, 103949, pp. 1-12.
- Sustainable Dundee's Urban ReLeaf Project
- The State of Nature Report (2019)
- Third State of Scotland's Greenspace report (2018)
- Water environment, open space, play, recreation and sport (2024)
- Ward, K., Beresford-Dey, M., Frediani, K. and Lakin, E. (2024) *NatureScot Research Report 1363 - Exploring engagement with Dundee Wee Forests and green spaces*

DATA QUALITY ISSUES

Dundee greenspace GIS data

- Issues could arise in naming conventions for *AreaName*, which could be inconsistent or not match with other datasets and may require normalisation before combining.
- One *AreaName* contains a comma, which could cause parsing issues if interpreted as a delimiter.

Personal well-being estimates data

- Only contains data for Dundee as a whole, doesn't distinguish between areas.
- Differing interpretations of the data; high happiness scores relate to a positive wellbeing, while high anxiety scores relate to a poor wellbeing.
- The dataset contains a title, so if importing it we would need to skip the title rows.
- This data is in wide form and will need to be converted to long form for time-series analysis.

Dundee social deprivation data

- Some population zones have varying population sizes, which could possibly distort the interpretation.

Play Area Sports and Recreation Playground location data

- Some rows contain unneeded spaces and line breaks.